

Vinay Kumar Ratnala

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EDUCATION

Stony Brook University

Master of Science in Data Science GPA: 3.6/4

Coursework: Algorithms, Operating Systems, Distributed Systems, Software Engineering

Stony Brook, NY

May 2025

Vellore Institute of Technology

Bachelor of Technology in Computer Science GPA: 9.1/10

Coursework: Data Structures, Database Management Systems, Object Oriented Design, Networks

Vellore, India

July 2023

EXPERIENCE

AI Innovation Institute

Research Assistant

Stony Brook, NY

September 2024 - Present

- Designed and deployed scalable, fault-tolerant inference services for RT-1 Transformer using GKE (Google Kubernetes Engine) on linux-based environments, aligned with distributed computing and Agile principles for educational robotics research.
- Implemented CI/CD workflows for continuous deployment tied to version control, enabling agile and iterative development.
- Engineered TensorFlow parsers and quaternion-to-trajectory converters to visualize robot motion in 3D—enhancing predictive analytics capabilities.

YiPi Bear

Software Engineer

Remote

May 2024 - Present

- Spearheaded development of an AI-driven full-stack platform using LangChain, FastAPI, MongoDB, and React to manage end-to-end user interactions in a learning-focused environment.
- Integrated secure JWT-based authentication and scalable RESTful APIs; built and deployed services using Docker, Prometheus, and Kubernetes.
- Applied software design principles to build modular and extensible microservices, enabling scalable feature deployment, improved code reusability, and simplified maintenance across distributed environments.

Dorklab

SDE Intern

Remote

December 2021 - February 2022

- Collaborated cross-functionally in Agile sprints to debug critical issues and deliver production-ready microservices under tight timelines using SpringBoot, Kafka, and PostgreSQL.

SKILLS

Programming Languages: Java, C++, Python, JavaScript, R, SQL, HTML, CSS, Scala

Tools & Frameworks: AWS, GCP, Docker, Kubernetes, Flask, FastAPI, Django, React, Angular, MongoDB, PostgreSQL, TensorFlow, PyTorch, CI/CD (GitHub Actions, Jenkins), Kafka, Elasticsearch, Linux, Oracle.

Concepts: Object-Oriented Design, Distributed Systems, Agile Development, Optimization, Educational Technology.

RELEVANT PROJECTS

Interactive Storytelling Platform

- Developed a scalable, full-stack storytelling platform integrating LLAMA and Waifu Diffusion through containerized microservices, enhancing interactive narrative learning experiences.
- Reduced response latency by 40% via prompt caching and asynchronous processing, enabling real-time narrative generation.
- Designed genre-specific prompt-chaining algorithms to generate branching stories, optimizing user experience

ZoneClock

- Engineered a Chrome extension that dynamically displays real-time local timestamps next to WhatsApp Web contacts, improving cross-timezone communication for global users.
- Leveraged JavaScript and phone-number-to-timezone API to dynamically convert phone numbers to geo-localized timestamps with 95% accuracy.
- Optimized UI responsiveness by implementing MutationObservers and Chrome APIs to monitor and update DOM changes, improving update speed by 30% in dynamic content environments.

Moviepedia

- Developed front-end interfaces with React.js, incorporating JWT authentication to personalize content based on user watch history. Engineered database architecture and implemented CRUD operations for 5 entities using Node.js.

aRAGorn

- Created a chatbot using GPT-3.5 and BM25 for contextual argument generation from large structured data (Kialo).
- Built prompt chaining with few-shot learning and chain-of-thought reasoning to tackle ambiguous input scenarios effectively.

Careers Website

- Created a job portal with location-based search and one-click apply, integrating Google Maps API and Flask-based backend.
- Reduced average page load time by 35% via frontend optimizations and backend query tuning.
- Integrated Google location services to offer nearby job suggestions and implemented profile-based email notifications for relevant job opportunities, enabling one-click applications.

ACHIEVEMENTS

- Placed 4th out of 1500 participants in a hackathon organized by VIT SAS. - 2019.
- Secured first place in the HackRoxx Hackathon at Vellore Institute of Technology, surpassing over 150 teams.